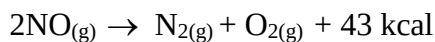
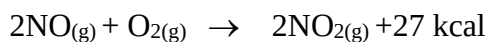


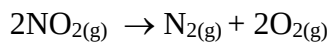
REACTION HEAT

1. Give three examples to exothermic and endothermic reactions.
2. Identify each of the following reactions as exothermic or endothermic.
 - a) $\text{H}_2\text{O}_{(g)} + 57,8 \text{ kcal} \rightarrow \text{H}_{2(g)} + 1/2\text{O}_{2(g)}$
 - b) $\text{Zn}_{(s)} + 2\text{HCl}_{(g)} \rightarrow \text{ZnCl}_{2(s)} + \text{H}_{2(g)} + 55,2 \text{ kcal}$
 - c) $\text{N}_{2(g)} + \text{O}_{2(g)} \rightarrow 2\text{NO}_{(g)} \quad \Delta H = +43,2 \text{ kcal}$
3. The molar heat of combustion of liquid ethyl alcohol to form carbon dioxide and water, is 295 kcal.
 - a) Write a balanced equation for the reaction.
 - b) How much heat, in kcal, is liberated when 9,2 g of alcohol are burned?
4. A 212 kcal of heat is liberated from the combustion of one mole of methane. How many grams of methane must be burned to increase the temperature of 1 kg of water from 15°C to 95°C?
5. Given the equation; $\text{N}_{2(g)} + 3\text{H}_{2(g)} \rightarrow 2\text{NH}_{3(g)} \quad (\Delta H = -22 \text{ kcal})$
Calculate the heat liberated in the formation of 85 g of ammonia.
6. When 0,4 mole of ethyl alcohol is burned, 118 kcal are liberated. If $\Delta H^0_f (\text{H}_2\text{O}) = -57,8 \text{ kcal}$, and $\Delta H^0_f (\text{CO}_2) = -94 \text{ kcal}$, what is the enthalpy of formation of ethyl alcohol?
7. Given; H-H : 104 kcal/mole, H-O : 110,8 kcal/mole and O=O : 118,2 kcal/mole
What is the molar enthalpy of formation of water?

8. Given the reactions;



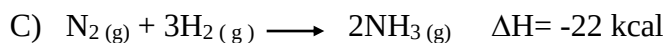
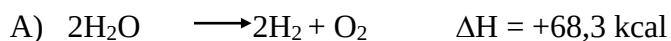
What is the enthalpy (ΔH) of the following reaction?



9. Which one(s) of the following is (are) endothermic?

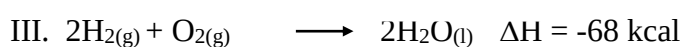
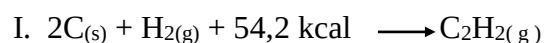
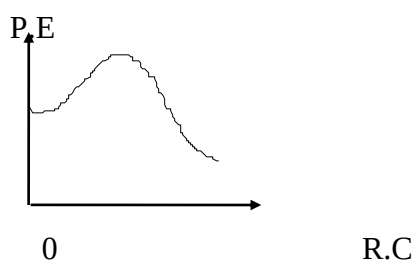
- I. Evaporation of water
- II. Electrolysis of water
- III. Burning of wood

10. Which one of the following reactions is an exothermic reaction ?



E) Melting of ice

11. Which one(s) of the following reaction(s) could be represented by the given PE-RC diagram ?



12. The complete combustion of carbon is represented by the equation $C_{(s)} + O_{2(g)} \longrightarrow CO_2 + 94 \text{ kcal}$.

How much heat is evolved in the complete combustion of 2,4 gram carbon ?

13. Given the reaction;



What is the enthalpy of formation of NO in kcal/mol ?



To obtain 188 kcal of heat , how many grams of sulfur must be burned ?

15. Given the following enthalpies of some Compounds;

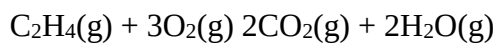
Compound Enthalpy of Formation

$C_2H_4(g)$ 12 kcal/mol

$CO_2(g)$ -94 kcal/mol

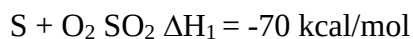
$H_2O(g)$ -68 kcal/mol

For the reaction;



Calculate the molar heat of combustion(ΔH) for that reaction ?

16. Given the following reactions with their enthalpies;



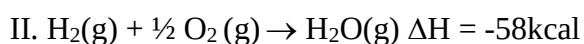
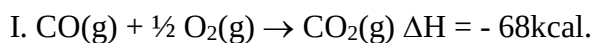
For the reaction; $\text{S} + 3/2\text{O}_2 \rightarrow \text{SO}_3$ What is ΔH value ?

17. 5,6 g of iron react with excess oxygen to produce (iron III oxide), Fe_2O_3 and 19 kcal of heat are released in this reaction. What is the enthalpy of formation of Fe_2O_3 in kcal ?

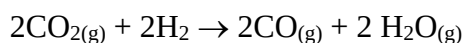
18. Given the reaction; $\text{H}_{2(g)} + \text{I}_{2(g)} \rightarrow 2\text{HI} + a \text{ kcal}$.

What is the enthalpy of formation of HI?

19.

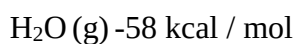
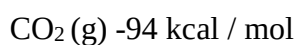
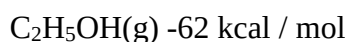


What is the ΔH for the following reaction in kcal ?



20. Given that;

Compound Enthalpy of formation



How many kcal of energy will be produced by the burning of 23g $\text{C}_2\text{H}_5\text{OH}$?